

Guruprasad

Analog Design Engineer

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Professional Summary

Analog Design Engineer with hands-on experience in LDO and power-management circuits in TI CMOS processes. Worked on Texas Instruments client projects involving block-level LDO design support and simulation, PVT and Monte-Carlo analysis, trimming, aging, and post-layout simulations. Experienced in silicon/IP-level characterization of non-volatile memories, VCO, HVGEN, and EEPROM IPs, including testbench development without data sheets. Strong background in analog fundamentals, PMIC blocks, and Cadence Spectre-based design flow.

Core skills

Analog Blocks: LDO, Bandgap, Op-Amps, Comparators, POR, Voltage References

PMIC Concepts: Stability, PSRR, Load Transient, Compensation, Trimming

Verification: PVT, Monte-Carlo, Aging, Burn-in, Corner Analysis

Simulation & Analysis: AC, Transient, Noise, Stability analysis

EDA Tools: Cadence Virtuoso, Spectre, ADE

Technology: TI pdks

Texas Instruments : Client Projects (via KarMic Design)

1. NMOS LDO Design

- Bias circuit modification and block-level design support and verification of NMOS LDO.
- PVT, Monte-Carlo, aging, and burn-in simulations
- Line/load regulation, PSRR, stability, startup, and load transient analysis
- Resistor trimming and post-layout simulation support

2. Embedded Memory & NVM IP Characterization

- Worked on EEPROM, MTP, and NVM IPs across multiple TI technologies.
- Developed testbenches for IDDQ, program/read currents, margin testing.
- External IREF margin characterization across PVT and Monte-Carlo (up to 1000 samples per corner)
- Post-burn and post-bake current analysis and data extraction.

3. High-Voltage Generator & VCO IP Testing

- HVGENMINILV / HVGENMINIMVAL IP characterization.
- Designed testbenches to measure IDDQ, frequency vs control voltage.
- Verified VCO performance across temperature and process corners.
- Supported performance improvement through circuit-level modifications.

Analog and Mixed-Signal Design Projects (Training / Internal)

1. PMOS Capless LDO

- Designed and verified a PMOS cap-less LDO with focus on stability, quiescent current, and load regulation.
- Conducted PVT and Monte-Carlo simulations to verify all the performance parameters and compared with the theoretical values.

2. Bandgap Reference with trimming

- Designed and verified current mode sub 1V bandgap voltage reference using vertical PNP based PTAT, CTAT currents and high gain folded cascode operational amplifier.
- Implemented a 5 bit resistor trimming circuit to tune the BGR across Process corners and mismatch

3. Design and Implementation of Synchronous voltage mode CCM Buck Converter

- Designed a synchronous CCM buck converter comprising Type II compensated folded cascode amplifier, Comparator, Level shifter, Non-overlapping single phase clock generator, Power switch driver, Power switches.
- Verified each block and simulated Buck converter across PVT corners and measured performance parameters.

4. Design and Implementation of 4 bit SAR ADC

- Designed a SAR ADC comprising Sample and Hold, Comparator, Successive Approximation Logic, Output buffer, Digital to Analog converter, Internal clock and control signal generator.
- Verified its functional correctness across the PVT corners and measured essential performance parameters.

5. Design and Implementation of Delta Sigma Modulator

- Designed a Delta Sigma Modulator comprising Fully differential folded cascode op-amp with common mode feedback, First order differential switched capacitor active integrator, Dynamic clocked comparator, One bit Digital to Analog converter, Non-overlapping two phase clock generator.
- Verified each block and simulated the modulator across PVT corners and measured vital parameters.

6. Design and verification of Digital Phase Locked Loop

- Designed a Digital PLL comprising Phase frequency Detector, Charge pump, Loop filter, Current starved 3 stage VCO, Divide by 2.
- Verified across all PVT corners and ran Monte-Carlo simulation.

7. Bandgap Reference in SKY130 - Tiny Tapeout Submission

- Set up an open-source analog design flow using Xschem, ngspice, Magic, Netgen, and the SkyWater SKY130 PDK.
- Designed and verified a Bandgap Reference with a reference voltage of approximately 0.9 V through PVT corner and Monte-Carlo simulations.
- Completed physical design, DRC, LVS, RC extraction, and post-layout verification.
- Submitted the BGR design to Tiny Tapeout for fabrication using the SKY130 process; silicon is expected in December 2026.

8. CMOS Comparator in SKY130 - Submission Ready

- Designed and verified a CMOS comparator using the SkyWater SKY130 PDK.
- Completed schematic design, layout, DRC, LVS, RC extraction, and post-layout simulation.
- Prepared the design as a submission-ready analog block following the Tiny Tapeout/SKY130 physical design and verification flow.

Education

B.E - Electronics and Communication

M.Tech - Digital Electronics and Communication

Ph.D - Analog Circuits

Previous Work Experience

- Design Engineer - KarMic Design Pvt. Ltd. Manipal. (Nov 2024 - Present)
- Teaching Faculty at MIT Manipal and SMVITM Bantakal - ECE Dept. (2008 - 2024)